Midterm 1

● Graded

Student

Marcus Huang

Total Points

102.5 / 150 pts

Question 1

1a

10 / 10 pts

- ✔ **– 0 pts** Correct and clear reasoning: has economic logic, typically related to social welfare, equity, diminishing marginal utility
- **– 4 pts** Correct but less clear explanation: some economic logic, incorrectly refers to redistribution as response not reason
- **– 6 pts** minimal economic logic applied, but attempted
- **– 8 pts** largely blank, no or incorrect economic logic
- **– 0 pts** Come back to

Question 2

1b

10 / 10 pts

- ✔ **– 0 pts** Fully correct explanation
- **– 3 pts** Partial credit
- **– 5 pts** Partial credit
- **– 7 pts** Partial credit
- **– 10 pts** Explanation not sufficient for credit

Question 3

1c

10 / 10 pts

- ✔ **– 0 pts** full credit
- **– 4 pts** almost there, typically restates the statement without additional reasoning. provides formula without economic intuition
- **– 6 pts** confusing answer
- **– 8 pts** no explanation

Question 4

1d **0** / 10 pts

- **− 0 pts** Correct
- **− 1 pt** Partial credit, typically misses point about assuming same utility functions or explanation not quite perfect
- **− 3 pts** Partial credit, typically misses point about assuming same utility functions and misses equal marginal utility implies equal consumption under this assumption
- **− 5 pts** Partial credit
- **− 7 pts** Partial credit
- ✔ **− 10 pts** Insufficient for credit to be awarded

Question 5

1e **6** / 10 pts

- ✔ **+ 2 pts** Correctly notes that the statement is false
- **+ 8 pts** Thorough and complete explanation
- **+ 6 pts** Solid explanation
- ✔ **+ 4 pts** Sufficient explanation
- **+ 2 pts** Limited explanation
- **+ 0 pts** Blank or incorrect without explanation

Question 6

2a **9.5** / 12.5 pts

- **− 0 pts** Student says it is not good causal evidence **AND** explicitly explains that the difference could be driven by pre-existing differences
- ✔ **− 3 pts** Student correctly says it is not causal evidence **BUT** reasoning is incomplete or slightly off. More specifically, they do not explain how baseline differences could explain the observed difference.

  Common example: students who jump straight to difference-in-differences and/or parallel trends, which is not required here for full credit.
- **− 6 pts** Student correctly says it is not causal evidence **BUT** reasoning is either completely off, unclear, or weak
- **− 9.5 pts** Answer is incorrect but makes some attempt
- **− 12.5 pts** blank

Question 7

2b

12.5 / 12.5 pts

✔ **− 0 pts** Yes, the parallel trends assumption needed for a difference-in-differences design appears to hold. A clear gap opens after the tax policy indicating an increase in home sales among the 125-175K price range

− 3 pts almost there. does not make reference to the gap after the policy change, signaling an increase in home sales. parallel trends allows you to use diff-in-diff, but whether the effect is positive, negative, or zero depends on post-period, too. full credit makes note of both the pre- and post-periods.

− 6 pts not quite right. typically focused on other factors, like the recession, only. yes, they are both decreasing in part because of the recession. in the absence of the tax break, they both would have fallen similarly. however, here, the treated group is either increasing or not falling by as much as the control, suggesting the tax rate lead to an increase.

− 9 pts wrong but attempt was made

− 12.5 pts blank

Question 8

2c

12.5 / 12.5 pts

✔ **− 0 pts** Correct answer of +50% with correct working

− 4.5 pts Gives correct formula but then computes incorrectly

− 8 pts Finds the changes for both treatment and control or gets -50%

− 12.5 pts None of the above apply

Question 9

2d

11 / 12.5 pts

+ 12.5 pts Complete answer that notes connection to parallel trends or potential confounding variables and explains why this evidence is reassuring

✔ **+ 11 pts** Strong answer that notes connection to parallel trends or potential confounding variables

+ 10 pts Strong answer overall, but could expand more on the connection to parallel trends assumption or the potential for unobserved differences

+ 10 pts Strong answer overall, but explanation not fully accurate

+ 8 pts Correct but needed more explanation

+ 6 pts Incomplete or incorrect explanation

+ 3 pts Very limited explanation

+ 1 pt Correct without explanation

+ 0 pts Blank / No explanation

Question 10

3a

10 / 10 pts

✔ **− 0 pts** Correct

− 3 pts One price or quantity correct

− 6 pts Sets supply=demand but gets both price and quantity incorrect

− 10 pts Does not set supply equal to demand or presents multiple methods one of which is incorrect

Question 11

3b

1 / 10 pts

+ 10 pts perfect answer
correct set up, where $Q_D = 912 - \frac{p+60}{5} = \frac{p}{10} = Q_S$
consumer price = 3060
producer price = 3000
quantity = 300

+ 9 pts correct set up **but** wrong math leading to one wrong value

+ 8 pts correct set up **but** wrong math leading to two wrong values

+ 7 pts correct set up **but** wrong math leading to three wrong values

+ 3 pts set up was wrong but involved solving for the equilibrium, typically solving for
$Q_D = 912 - \frac{p-60}{5} = \frac{p}{10} = Q_S$ **or**
$Q_D = 912 - \frac{p}{5} + 60 = \frac{p}{10} = Q_S$ **or**
$Q_D = 912 - \frac{p}{5} - 60 = \frac{p}{10} = Q_S$ **or**
$Q_D = 912 - \frac{p+60}{5} = \frac{p-60}{10} = Q_S$ **or**
some other iteration of this

✔ **+ 1 pt** set up was wrong, typically just adding 60 to the equilibrium price from part a

+ 0 pts blank

Question 12

3c

6 / 10 pts

+ 5 pts DWL computation correct

✔ **+ 2.5 pts** DWL computation present but incorrect or incomplete

+ 0 pts DWL computation missing

+ 5 pts Supply-demand graph correct and labelled

✔ **+ 2.5 pts** Supply-demand graph present but missing key elements or mislabelled DWL

+ 0 pts Supply-demand graph missing

✔ **+ 1 pt** Flex point

Question 13

3d

1.5 / 10 pts

Incidence

+ 0 pts Click here to replace this description.

+ 5 pts Correct tax incidence calculations

+ 2.5 pts Incomplete or incorrect incidence calculation

✔ **+ 0 pts** Missing tax incidence calculation

Intuition

+ 0 pts Click here to replace this description.

+ 2 pts Correctly argues that producers bear more of the burden

+ 3 pts Explains that producers bear more of the burden because supply is less elastic than demand

✔ **+ 1.5 pts** Incomplete or incorrect explanation of intuition behind incidence

+ 0 pts Missing intuition behind incidence

+ 1 pt Flex point

Question 14

3e

2.5 / 10 pts

− 0 pts Correctly states that this means no deadweight loss due to no quantity change and that the consumers bear the full incidence through the tax-inclusive price

− 2.5 pts Both conclusions correct but only one is fully explained

− 5 pts One conclusion incorrect or both conclusions correct but not fully explained

✔ **− 7.5 pts** One conclusion correct and both not fully explained

− 10 pts Incorrect, including some correct conclusions with incorrect reasoning

Econ 131
Spring 2026

Midterm

March 2026

Student Name: Marcus Huang

Student ID: 3041432436

Exam Instructions

- **This exam has three questions. Each question is worth 50 points (150 points total). The total exam time is 75 minutes.**
- **Write answers using pens (we recommend black or blue ink), and write only in the space provided for each answer so your solutions scan properly.**
- **Do not write your solutions on pages that say "Do not write on this page".** Answers written on these pages will not be graded.
- **Show your work.** Credit will only be awarded on the basis of what is written on the exam.
- **Affirm the academic honesty pledge below.** For those writing on a non-printed copy, please write "Academic Honesty Pledge as on exam" and sign your name. **If you do not affirm this pledge, your exam will be marked <u>invalid</u>.**

0. ACADEMIC HONESTY PLEDGE

I confirm that I have abided by all academic honesty rules for UC Berkeley and Economics 131. I confirm that I did not see this exam before my official exam start time. I confirm that I have not shared and will not share this exam with anyone else.

Signature: Marcus Huang

1. True/False/Uncertain Questions (50 points total)

You can start your answer "True", "False", "Uncertain", "Partially true", or "Partially false", etc., but your choice of the label does not matter much. What matters is your explanation.

(a) Redistribution is one reason for government intervention in the economy. *(10 points)*

Yes, it is true. The main reasons for government intervention are:
① Market failure and ② Redistribution

The market or economy will reach a balanced point even one person gets everything while others get nothing. That's sort of the gap between the rich and the poor. redistribution make the distribution of money more fair. making sure the economy of the society not collapse.

(b) In a simple observational correlation, a large sample is enough to make the estimate causal. *(10 points)*

This statement is not true. Only having a large sample is not enough. Estimate causal should consider all the things comprehensively. Avoiding measurement bias, the group of sample we choose are important to the estimation as well. otherwise the correlation might be misleading.
Then the unit we use to measure is important as well.
We should make sure less bias, while we can not eliminate variety/diversity.

$g > r$

(c) If the GDP growth rate is higher than the interest rate on government debt, then the government can run a primary deficit while keeping debt/GDP stable. *(10 points)*

Yes, this is true. the g (GDP growth rate) and r (interest rate on debt) are related. As we know, to make a sustainable debting system we ~~shon~~ must keep $r < g$. then the input and output of the economy can be balanced. In that way, ~~the debt~~ ~~we~~ the government can run a primary deficit while keeping debt/GDP stable.

(d) If taxes and transfers have absolutely no behavioral effects, a utilitarian government sets taxes and transfers so that everyone consumes the same amount. *(10 points)*

This statement is not True to me. The consumers consume the goods based on their preferences. And if there will be no behavioral effects on taxes and transfer, which ~~men~~ means ~~their~~ they are inelastic, they will still kept the previous consuming habits. That is vary from people. they might not consume the same amount of goods. We can say they consume as same as themselves previously. pre-taxed and no-transfer.

EITC

(e) An Earned Income Tax Credit is desirable when the intensive margin response is larger than the extensive margin response. *(10 points)*

This statement is false to me.

In my own understanding, intensive marginal response more means the people are elastic. And the tax should always focus on inelastic perspective to maximize the efficiency. thus, for an elestic system/situation right here, the EITC is not that desirable to appear.

But the EITC techically is a tool to redistribute the money allocation. The poor people are taxed less with larger EITC and middle class ~~have~~ has a flat range, while the rich people are taxed more based on their incomes.

2. Empirical Application: Best and Kleven (Housing Market Responses to Transaction Taxes) (50 points total)

In the the United Kingdom, the government taxes home sales. In resonse to the Great Recession, the UK government in Setptember 2008 eliminated the tax on homes in the sale price range between £125,000 and £175,000. This temporary tax break lasted for 16 months, from September 2008 through December 2009. The tax rate on properties in the sale price range between £175,000 and £225,000 remained unchanged.

For all figures in this question, the y-axis ranges from -125% ***to*** 50%.

(a) Suppose you find that in the 16-month period, more homes were sold in the £125,000-£175,000 price range than in the £175,000-£225,000 price range. Is that good evidence that the tax break caused an increase in home sales in the £125,000-£175,000 price range, why or why not? *(12.5 points)*

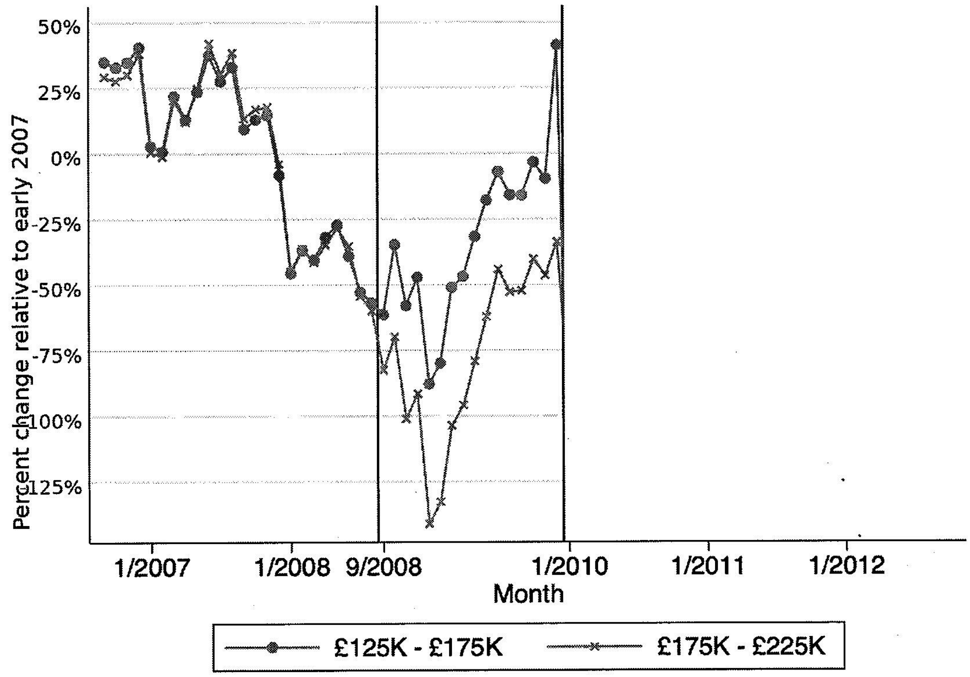
Short answer : maybe (mostly not ~~enough~~)

When talking about causal relationship, we can use Difference - in - differences ~~motho~~ method to see if there is a relationship between them.

According to the question and info, we can't actually know the trend ~~when~~ before, with the abscence of the policy.

$$DiD = (Y^{t,post} - Y^{c,post}) - (Y^{t,pre} - Y^{c,pre})$$

~~we can not~~ the sales ~~do~~ did increase. but is it absolutely related to the implement of the policy (tax break)? we can not make sure. but this is the part $(Y^{t,post} - Y^{c,post})$ so it can be an ~~good~~ evidence ~~perha~~ perhaps. but we can not make conclusion only based on this information alone.



(b) Suppose you draw the figure above. The vertical black line indicates September 2008. The circles show the number of home sales in the £125,000-£175,000 price range, expressed as the percent change from early 2007. The stars show the number of home sales in the £175,000-£225,000 price range, also expressed as the percent change from early 2007. Is this good evidence that the tax break caused an increase in home sales in the £125,000-£175,000 price range, why or why not? *(12.5 points)*

Short answer : Yes !

This diagram shows that when there was abscence of the policy, it was not start yet, these two range of prices's sales are in the same trend, very close. Then after the policy, they remain the similar trend, ~~m~~ I mean the shape of lines, while £125k-£175K is always above the £175k-£225k.

(c) Using at the graph above, approximately what do you think a difference-in-differences table would show is the difference-in-differences estimate of the causal effect of the tax break on home sales in the £125,000-£175,000 price range? (You may use the following numbers. The graph's pre-treatment mean was approximately 0% for both the £125,000-£175,000 group and for the £175,000-£225,000 group. The graph's post-treatment mean was approximately −25% for the £125,000-£175,000 group and approximately −75% for the £175,000-£225,000 group.) *(12.5 points)*

treatment group : £125k - £175k as t

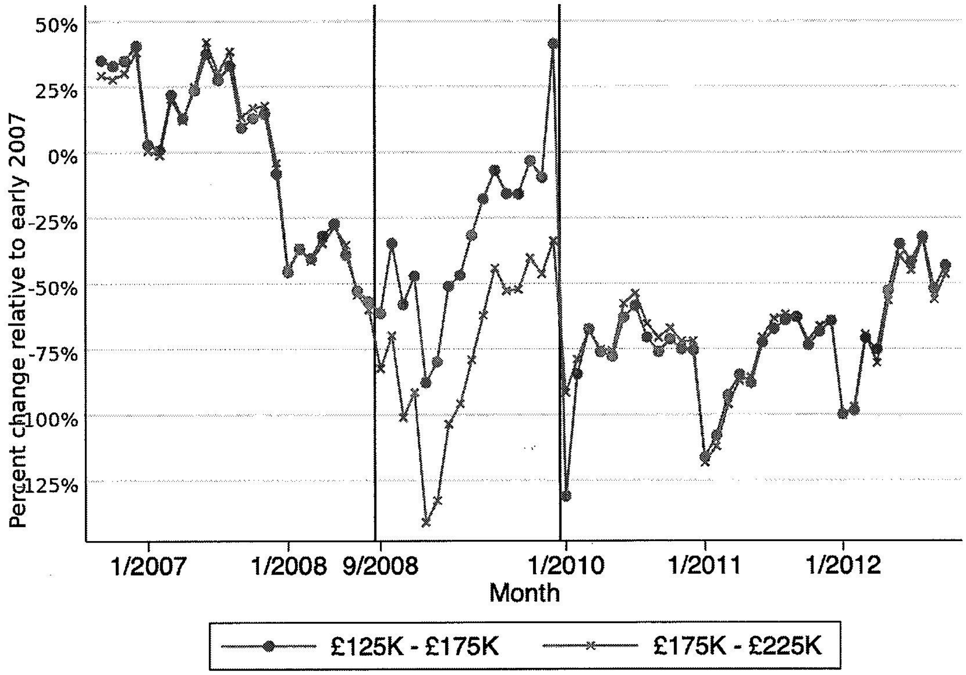
control group : £175k - £225k as c

$$DiD = (Y^{t,post} - Y^{c,post}) - (Y^{t,pre}, Y^{c,pre})$$

$$= ((-25\%) - (-75\%)) - (0\% - 0\%)$$

$$= -25\% + 75\%$$

$$= \underline{\underline{50\%}}$$



(d) Now suppose you extend the figure to include data points after December 2009, as in this second figure. Does this new figure provide additional comforting evidence that the tax break caused an increase in home sales in the £125,000-£175,000 price range, why or why not? *(12.5 points)*

Yes ! this graph shows clear that : before the tax ~~be~~ break, these two groups have the same (almost) trend of change. ~~after to~~ after implemented tax break, difference appeared, very clear. and still in same shape but different ~~amount~~ percentage change. After remove tax break, these two groups ~~go~~ went back to the same track, almost the same trend same change then. So this is a confoting evidence ~~$~~ to reinforce the change ~~eas~~ caused by the tax break, the relationship is ~~prof~~ ~~proofed~~ prooved. From now, the experimen is good circle ~~comptem~~ conpleted.

3. Incidence of Commodity Taxation (50 points total)

Consider the following stylized market for televisions in Berkeley. Let market demand and supply be summarized as

$$Q^D = 912 - \frac{P}{5}, \qquad Q^S = \frac{P}{10},$$

where P is the dollar price per TV and Q is measured in thousands of TVs.

(a) Solve for the competitive equilibrium before any tax. Report equilibrium price and quantity. *(10 points)*

Equilibrium appears when $Q^D = Q^S$. thus : $912 - \frac{P}{5} = \frac{P}{10} \Rightarrow 912 = \frac{P}{10} + \frac{2P}{10}$

$\Rightarrow \frac{3P}{10} = 912 \rightarrow 3P = 9120 \rightarrow P^* = 3040$ dollar/TV

the $Q^* = \frac{P}{10} = \frac{3040}{10} = \cancel{34}\ 304$ k TVs.

(b) Now impose a per-unit tax of $t = \$60$ on buyers. Solve for the post-tax equilibrium and report the consumer price, producer price, and quantity. *(10 points)*

Equilibrium now : $\cancel{912 - \frac{P}{5} = P+60}$

price for consumer : 3100 dollar/TV

price for producer : ~~3040~~ 2920 dollar/TV

Quantity = 292 k TVs.

$P' = 3040 + 60 = 3100$ \$/TV.

~~3100~~,

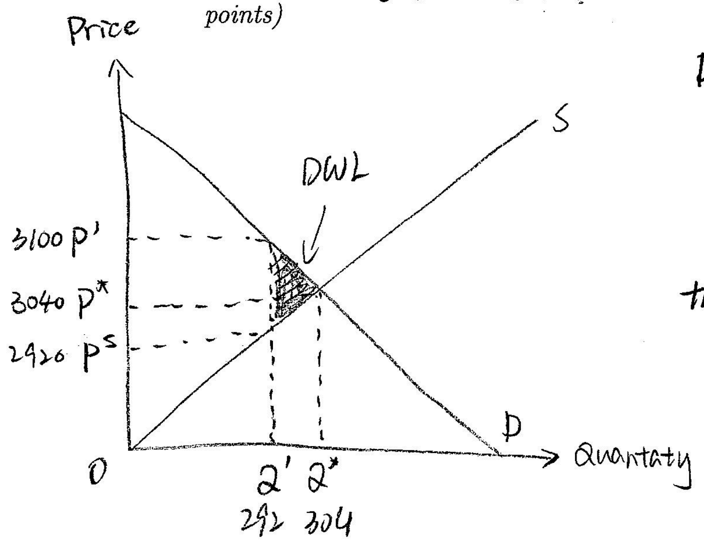
$Q' = 912 - \frac{3100}{5} = 912 - 620$

$= 292$ KTVs

$Q' = \frac{P^s}{10}$

$\Rightarrow 292 = 10P^s \rightarrow P^s = 2920$ \$/TV

(c) Compute the deadweight loss from the tax and illustrate it on a supply-demand graph. *(10 points)*



$$DWL = (3100-2920) \times (304-292) \times \frac{1}{2}$$
$$= 180 \times 12 \times \frac{1}{2}$$
$$= 180 \times 6$$
$$= 1080$$

the DWL is $1080

(d) Compute the tax incidence on consumers and producers, and explain intuitively which side bears more of the burden and why. *(10 points)*

The side which is inelastic burden more because they have no alternatives.

(e) Suppose consumers are fully inattentive to the tax and keep using the original demand equation as if no tax exists. What does this imply for deadweight loss and incidence? You do not need to do any math; full credit will be given for clear intuition. *(10 points)*

If the consumers ignore the tax and use the demand ~~quantaty~~ equation as before, the deadweight loss will goes up and they gonna burden more incidence. They are inelastic ~~de~~ due to not sensitive on the price change.

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